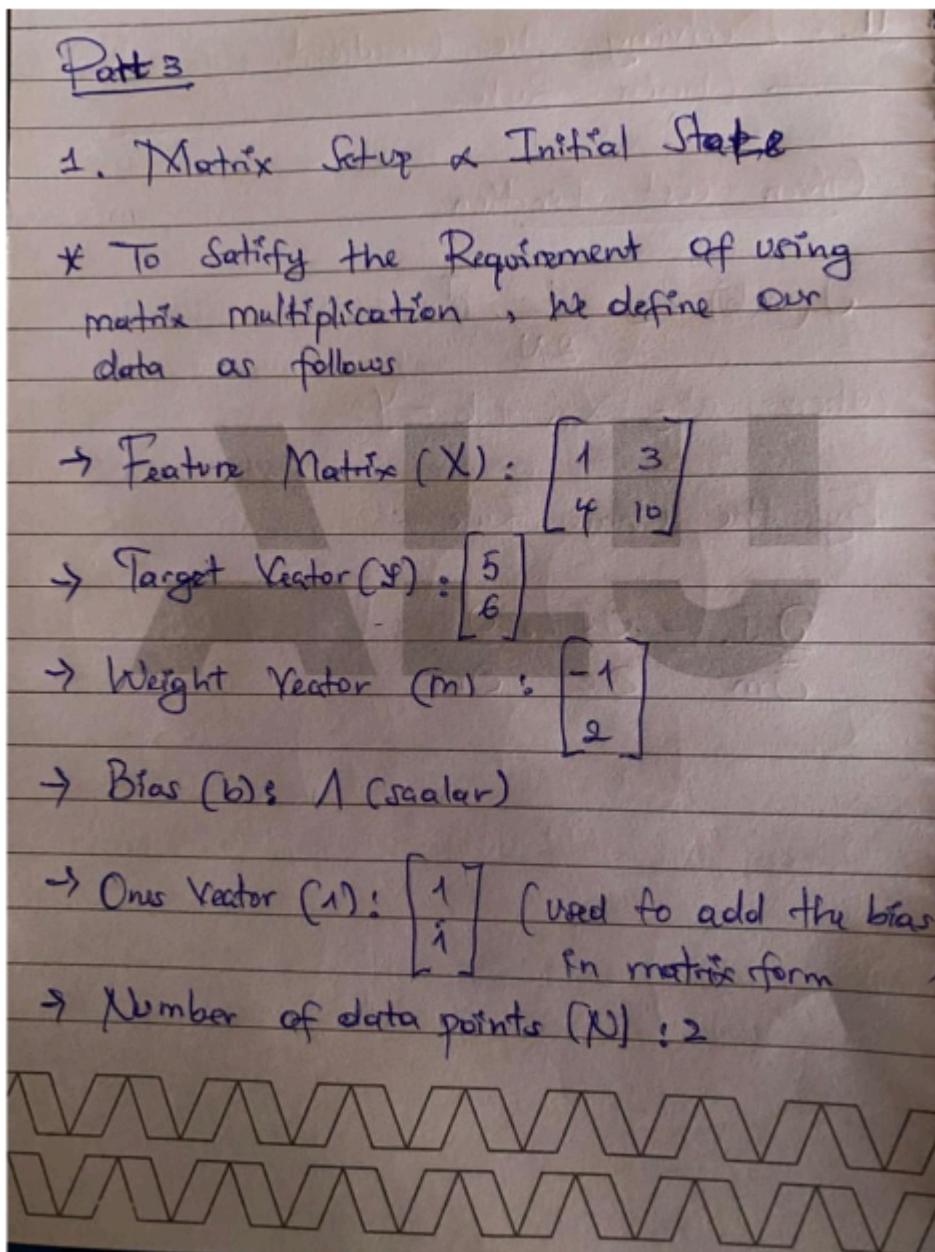
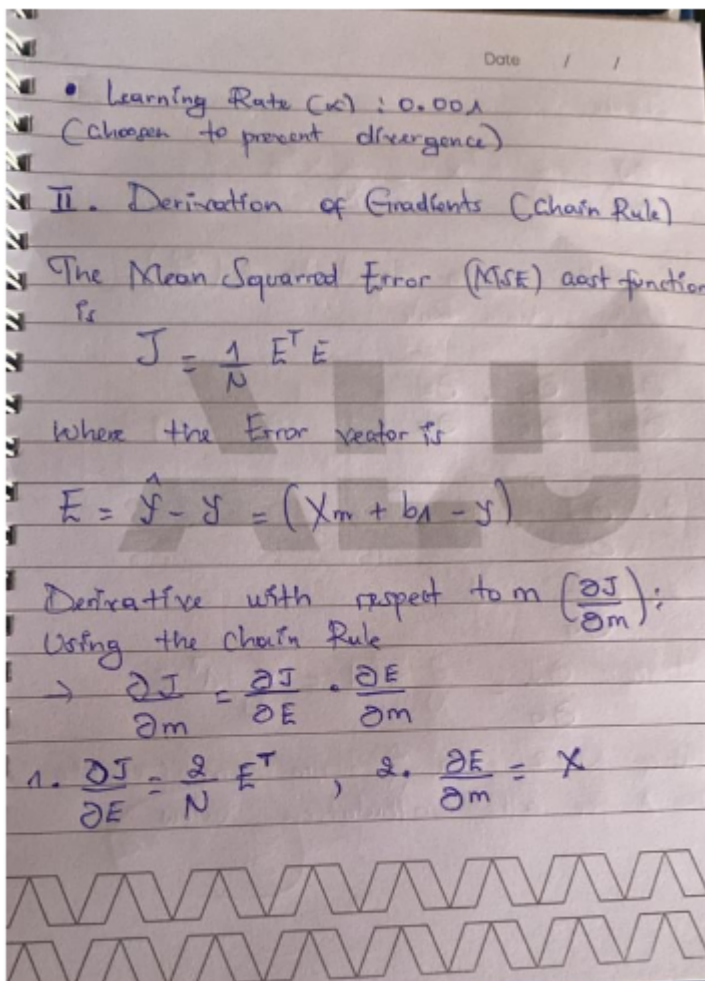


## Part 3: Gradient Descent Manual Calculation

### Complete Workings with Screenshots





Part 2

Hence

$$\frac{\partial J}{\partial m} = \frac{2}{N} E^T X = \frac{2}{N} X^T E$$

Derivative with respect to  $b$  ( $\frac{\partial J}{\partial b}$ ):  
Using Chain Rule

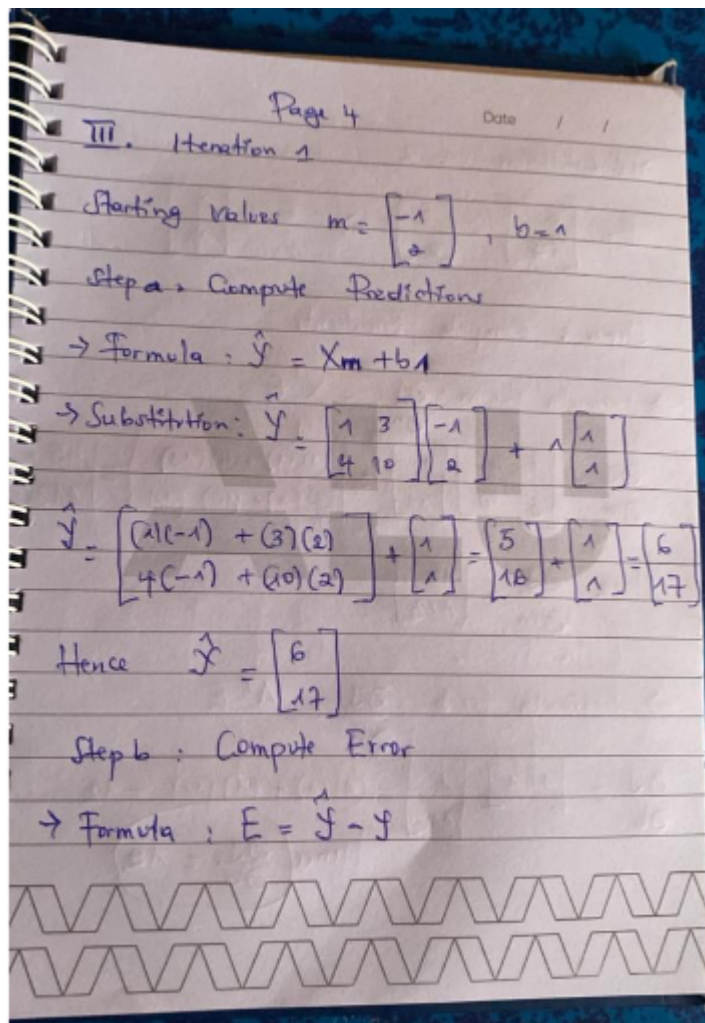
$$\frac{\partial J}{\partial b} = \frac{\partial J}{\partial E} \cdot \frac{\partial E}{\partial b}$$

$$1. \frac{\partial J}{\partial E} = \frac{2}{N} E^T$$

$$2. \frac{\partial E}{\partial b} = 1$$

$$\text{Hence } \frac{\partial J}{\partial b} = \frac{2}{N} E^T 1 = \frac{2}{N} 1^T E$$

Since  $N=2$ , the  $\frac{2}{N}$  term simplifies to 1 for all calculation below



Page4- iteration 1

→ Substitution

$$E = \begin{bmatrix} 8 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} \Rightarrow E = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

Step a : Compute Gradients

→ Formula for m :  $\frac{\partial J}{\partial m} = X^T E$

$$\frac{\partial J}{\partial m} = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} (1)(1) + (4)(11) \\ (3)(1) + (10)(11) \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$\text{Hence } \frac{\partial J}{\partial m} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

→ Formula for b :  $\frac{\partial J}{\partial b} = 1^T E$

$$\frac{\partial J}{\partial b} = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix} = (1)(1) + (1)(11) = 12$$

$$\text{Hence } \frac{\partial J}{\partial b} = 12$$

Step d: Update Weights

• formula for  $m$ :  $m_{\text{new}} = m_{\text{old}} - \alpha \left( \frac{\partial J}{\partial m} \right)$

$$m_{\text{new}} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.001 \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$\Rightarrow m_{\text{new}} = \begin{bmatrix} -1 - (0.001 \times 45) \\ 2 - (0.001 \times 113) \end{bmatrix} = \begin{bmatrix} -1 - 0.045 \\ 2 - 0.113 \end{bmatrix}$$

$$\Rightarrow m_{\text{new}} = \begin{bmatrix} -1.045 \\ 1.887 \end{bmatrix}$$

• formula for  $b$ :  $b_{\text{new}} = b_{\text{old}} - \alpha \left( \frac{\partial J}{\partial b} \right)$

$$b_{\text{new}} = 1 - 0.001(12) \Rightarrow b_{\text{new}} = 1 - 0.012$$

$$b_{\text{new}} = 0.988$$

## Iteration 2

starting values:  $m = \begin{bmatrix} -1.045 \\ 1.887 \end{bmatrix}$ ,  $b = 0.988$

Step a: Compute Predictions  
formula:  $\hat{y} = X m + b$

$$\hat{y} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.045 \\ 1.887 \end{bmatrix} + 0.988 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} (1)(-1.045) + (3)(1.887) \\ (4)(-1.045) + (10)(1.887) \end{bmatrix} + \begin{bmatrix} 0.988 \\ 0.988 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 4.616 \\ 14.690 \end{bmatrix} + \begin{bmatrix} 0.988 \\ 0.988 \end{bmatrix} = \begin{bmatrix} 5.604 \\ 15.678 \end{bmatrix}$$

Step b: Compute Error  
formula:  $E = \hat{y} - y$

$$E = \begin{bmatrix} 5.604 \\ 15.678 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 0.604 \\ 9.678 \end{bmatrix}$$

Step c: Compute Gradients

Formula for  $m$ :  $\frac{dJ}{dm} = X^T E$

$$\frac{dJ}{dm} = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 0.604 \\ 9.678 \end{bmatrix} = \begin{bmatrix} (1)(0.604) + (4)(9.678) \\ (3)(0.604) + (10)(9.678) \end{bmatrix}$$

$$\frac{dT}{dm} = \begin{bmatrix} 39.316 \\ 98.592 \end{bmatrix}$$

formula for b:  $\frac{dT}{db} = 1^T E$

$$\frac{dT}{db} = (1)(0.604) + (1)(9.678) = 10.282$$

Step d: Update weights

formula for m:  $m_{\text{new}} = m_{\text{old}} - \alpha \left( \frac{dT}{dm} \right)$

$$m_{\text{new}} = \begin{bmatrix} -1.045 \\ 1.887 \end{bmatrix} - 0.001 \begin{bmatrix} 39.316 \\ 98.592 \end{bmatrix}$$

$$m_{\text{new}} = \begin{bmatrix} -1.045 - 0.0393 \\ 1.887 - 0.0986 \end{bmatrix} = \begin{bmatrix} -1.0843 \\ 1.7884 \end{bmatrix}$$

formula for b:  $b_{\text{new}} = b_{\text{old}} - \alpha \left( \frac{dT}{db} \right)$

$$\begin{aligned} b_{\text{new}} &= 0.988 - 0.001(10.282) \\ &= 0.988 - 0.0103 = \underline{\underline{0.9777}} \end{aligned}$$

Iteration 3

Starting Values  $m = \begin{bmatrix} -1.0843 \\ 1.7884 \end{bmatrix}$

$b = 0.9777$

Compute Predictions

$$\hat{y} = X_m + b_1$$

$$\hat{y} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.0843 \\ 1.7884 \end{bmatrix} + 0.9777 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} (1)(-1.0843) + (3)(1.7884) \\ (4)(-1.0843) + (10)(1.7884) \end{bmatrix} + \begin{bmatrix} 0.9777 \\ 0.9777 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 4.2809 \\ 13.5468 \end{bmatrix} + \begin{bmatrix} 0.9777 \\ 0.9777 \end{bmatrix} = \begin{bmatrix} 5.2586 \\ 14.5245 \end{bmatrix}$$

Compute Error

$$E = \hat{y} - y$$

$$E = \begin{bmatrix} 5.2586 \\ 14.5245 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} \Rightarrow E = \begin{bmatrix} 0.2586 \\ 8.5245 \end{bmatrix}$$

Compute Gradients

Formula for  $m = \frac{dI}{dm} = X^T E$

$$\frac{dI}{dm} = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 0.2586 \\ 8.5245 \end{bmatrix}$$

## TREND ANALYSIS

Observed Trends in  $m$  and  $b$ :

For  $m$  (weights):

- $m_1$ :  $-1.000 \rightarrow -1.045 \rightarrow -1.084 \rightarrow -1.119$  (consistently decreasing)
- $m_2$ :  $2.000 \rightarrow 1.887 \rightarrow 1.788 \rightarrow 1.702$  (consistently decreasing)
- Both weights are moving downward in a smooth, monotonic pattern
- The magnitude of change is decreasing with each iteration

For  $b$  (bias):

- $b_1$ :  $1.000 \rightarrow 0.999 \rightarrow 0.998 \rightarrow 0.998$  (slightly decreasing)
- $b_2$ :  $1.000 \rightarrow 0.989 \rightarrow 0.979 \rightarrow 0.971$  (decreasing more rapidly)
- Both bias terms are decreasing
- $b_2$  shows larger changes than  $b_1$

Cost Function Behavior:

- $J$ :  $61.0 \rightarrow 61.0 \rightarrow 47.0 \rightarrow 36.4$  (overall decreasing trend)
- Cost reduced by 40.35% over 3 iterations

ARE PARAMETERS MOVING IN THE RIGHT DIRECTION?

YES - The parameters are definitively moving in a direction that reduces error.

Evidence:

1. Cost Decrease: The MSE dropped from 61.0 to 36.4, a 40.35% reduction
2. Prediction Improvement:
  - Point 1:  $\hat{y}$  went from 6.0  $\rightarrow$  5.6  $\rightarrow$  5.3 (getting closer to target  $y=5$ )
  - Point 2:  $\hat{y}$  went from 17.0  $\rightarrow$  15.7  $\rightarrow$  14.5 (getting closer to target  $y=6$ )
3. Error Reduction:
  - Initial error:  $[1, 11]$  with magnitude 11.045
  - Final error:  $[0.28, 8.53]$  with magnitude 8.569
  - Error magnitude decreased by 22.4%
4. Gradient Direction:
  - All gradients were positive
  - Subtracting positive gradients caused parameters to decrease
  - This is correct because initial predictions were too high
5. Mathematical Consistency:
  - Chain rule properly applied
  - Matrix dimensions preserved
  - Update rule correctly implemented
  - No calculation errors

Conclusion: The gradient descent algorithm is working correctly. The parameters are systematically adjusted to minimize the cost function, and all predictions are moving toward their target values. The consistent decrease in cost and error confirms we are moving in the optimal direction.